P11 Research Dossier

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Reading the two papers the project brief names settled four things P11 had been treating as unknown — most usefully, that GEO-bench ships full document text rather than URLs, and that the foundational GEO paper never attempts prediction at all, which is the gap this project fills.

What reading the papers changed

Four things the project had recorded as unknown or uncertain are now settled. Each came from reading a source rather than from the project brief or the project dossier.

Finding	What we had assumed	What the source says	Consequence
GEO-bench ships full text	Possibly only URLs, implying a scraping pipeline	"Each query is augmented with the cleaned text content of the top 5 search results from the Google search engine"	Removes a scraping and link-rot risk that was about to become a blocking question
GEO never predicts	Unclear how much novelty was left after reproduction	Zero occurrences of "predict" across 12 pages, against 51 of "visibility"	Predicting visibility *before* modification is not done in the foundational paper. This is P11's gap
Rank effects are large and non-monotonic	Rank was one feature among many	Cite Sources gave rank-5 sources **+115.1%** while the top-ranked source **lost 30.3%**	A rank-only control baseline becomes mandatory, not advisable
Liu et al. venue resolved	Brief said "Findings of the ACL"; dossier said EMNLP in one place, ACL in another	Findings of **EMNLP** 2023, arXiv:2304.09848	A citation discrepancy is closed before it reached a report

The caveat on novelty

"Zero occurrences of 'predict'" is a fact about *that paper*. It is not evidence that no follow-up work has built a predictive GEO model. A novelty check against post-2024 literature is still outstanding, and the claim should not be made in writing until it is done.

GEO (Aggarwal et al., KDD 2024)

The anchor paper: it defines the visibility metrics P11 must reproduce, supplies GEO-bench, and specifies the pipeline we are rebuilding. Read in full on 16 October 2026.

The two objective metrics

Both are directly implementable and need no LLM judge. `S_ci` is the set of sentences citing source `c_i`, `S_r` all sentences in the response, `|s|` the word count of sentence `s`. Where a sentence cites several sources, the word count is shared equally.

Word Count — the share of answer text attributed to a source:

```
\mathrm{Imp}_{wc}(c_i, r) = \frac{\sum_{s \in S_{c_i}} |s|}{\sum_{s \in S_r} |s|}
```

Position-Adjusted Word Count — the same, with earlier citations weighted more, on the grounds that click-through rates follow a power law in rank:

```
\mathrm{Imp}_{pwc}(c_i, r) = \frac{\sum_{s \in S_{c_i}} |s| \cdot e^{-\frac{\mathrm{pos}(s)}{|S|}}}{\sum_{s \in S_r} |s|}
```

A third metric, **Subjective Impression**, scores seven dimensions — relevance, influence, uniqueness, subjective position, subjective count, click likelihood and diversity — using G-Eval with GPT-3.5 as judge. It needs one judge call per citation per response, sampled repeatedly, so it is the project's largest cost risk and is treated as optional.

Visibility is continuous, not binary

This is the detail most likely to be missed. GEO's metric is a *share of answer text*, not a cited/not-cited flag. A source mentioned in one clause and a source carrying half the answer are both "cited" but score very differently. What P11 predicts is therefore an open design decision, not a given.

GEO-bench

10,000 queries, split 8,000 train / 1,000 validation / 1,000 test, drawn from nine sources: MS MARCO, ORCAS-1, Natural Questions, AllSouls, LIMA, Davinci-Debate, Perplexity.ai Discover, ELI-5, and GPT-4-generated queries. Spans 25 domains and 7 tag categories, preserving a real-world intent mix of 80% informational and 10% each transactional and navigational.

Each query carries the cleaned full text of the top 5 Google results.

The pipeline to reproduce

Component	Setting
Retrieval	Top 5 Google results per query
Source handling	No summarisation; whole source text in context
Generator	`gpt-3.5-turbo`
Sampling	Temperature 0.7, 5 sampled responses
Repetition	Averaged over 5 random seeds

The sampling protocol is itself evidence: the authors repeat every condition five times across five seeds because citation is stochastic. That is the empirical case for measuring our own label stability early.

`gpt-3.5-turbo` is deprecated, so a faithful reproduction needs an agreed substitute — a question for the supervisor, not a silent choice.

Stated limitations

The authors note that methods may decay as engines evolve, that query distributions shift, that they did not evaluate effects on conventional search rankings, and that tagging may contain errors. There is no ethics or broader-impact section — the paper runs 1, 2, 3, 4, 5, 7, 8, 9, 10, with no such section at all.

Liu et al. — citation is not support

The brief's second named paper audits four deployed generative search engines and finds that being cited and being supported are different properties. That distinction matters directly for how P11 reads its own labels.

The paper defines two metrics that are now standard vocabulary: **citation recall** (all statements are fully supported by citations) and **citation precision** (every citation supports its associated statement). Human evaluation covered Bing Chat, NeevaAI, Perplexity.ai and YouChat.

Measure	Result
Generated sentences fully supported by citations	**51.5%**
Citations that support their associated sentence	**74.5%**

The authors' framing is that responses "are fluent and appear informative, but frequently contain unsupported statements and inaccurate citations" — perceived quality and actual grounding come apart.

Why this constrains the project

P11's pipeline produces a binary `cited` label. This paper shows roughly a quarter of real citations do not support what they are attached to. So `cited = 1` is a **visibility signal, not a correctness signal**, and a model trained on it will learn whatever drives citation, including whatever drives spurious citation.

Two consequences follow. A content-optimisation system could raise visibility while degrading answer quality, which is the concrete basis for the dual-use limitations section the brief does not mention. And measuring citation precision and recall on our own generated answers is a cheap sanity check on label quality before those labels become classifier targets.

> **Status: read from the abstract and verified metadata only.** Venue, authors, arXiv ID and the two percentages above are confirmed. The methodology — annotator counts, query counts, rating scales — is unverified and must not be cited until the full paper is read.

Supporting literature

Six papers beyond the two the brief names. ALCE is the one that matters most and was promoted to high priority, because it specifies the citation-parsing step that manufactures every label in the project.

Paper	What we take from it	Priority
Gao et al., ALCE (EMNLP 2023)	The citation-parsing and labelling procedure, ready-made: segment by sentence, cap at 3 citations per statement, concatenate multiple citations and judge them jointly with an NLI model. Adopting it makes our citation-quality numbers comparable to published work	**High**
Lewis et al., RAG (NeurIPS 2020)	The canonical retrieve-then-generate citation. Its fixed-index design justifies treating the retriever as frozen while studying downstream behaviour	Medium
Thakur et al., BEIR (NeurIPS D&B 2021)	How to report the retrieval stage: strictly zero-shot, standardised on nDCG@10	Medium
Karpukhin et al., DPR (EMNLP 2020)	Dual-encoder dense retrieval; 9–19% over BM25 at top-20. Motivates similarity-score features	Medium
Nogueira & Cho (2019)	BERT cross-encoder re-ranking, +27% relative MRR@10. The natural advanced model if we compare transformers	Low
Burges, LambdaMART (MSR-TR-2010-82)	The learning-to-rank lineage our interpretable model sits in	Low

The low-priority two are gated on choosing the relevant advanced direction, which the brief says happens after first experimental results.

Open decisions this raises

Five questions that reading the papers created or sharpened. None can be answered from the brief, and the first determines the dataset schema, so it should be settled before anything is built.

1. **What exactly do we predict?** The brief says "selected and/or cited", which does not settle it. Three candidates: binary `cited`; continuous `Imp_pwc`; or a four-way class — not retrieved, retrieved-not-cited, cited-low-visibility, cited-high-visibility. Each implies a different schema, metric set and loss function.
2. **Which model replaces `gpt-3.5-turbo`?** It is deprecated, and the substitute will change the results we are trying to reproduce.

3. **Do we reproduce Subjective Impression?** It needs a G-Eval judge call per citation per response. Against a budget of roughly $500, this is plausibly the largest single cost line, while the two objective metrics need no judge at all.
4. **Do we condition explicitly on retrieval rank?** The +115.1% / −30.3% asymmetry says position effects are large enough that ignoring them would be a modelling error.
5. **Is the novelty predicting visibility before modification?** Worth confirming rather than assuming, and worth pairing with the post-2024 novelty check.

A fuller agenda, including budget mechanics, the December midterm and the dual-use question, lives in the repository at `docs/meeting-notes/2026-10-16-kickoff-agenda.md`.

Sources

Every field below was confirmed from a page opened on 16 October 2026. Fields that could not be confirmed are marked unverified and must not be cited until checked — the project's standing rule is that no bibliographic detail is written down unless it was read.

Paper	Venue	Identifier	Status
GEO: Generative Engine Optimization — Aggarwal, Murahari, Rajpurohit, Kalyan, Narasimhan, Deshpande	KDD 2024, Barcelona	arXiv:2311.09735	Verified; read in full
Evaluating Verifiability in Generative Search Engines — Liu, Zhang, Liang	Findings of EMNLP 2023	arXiv:2304.09848	Verified; methodology unread
Enabling LLMs to Generate Text with Citations — Gao, Yen, Yu, Chen	EMNLP 2023, pp. 6465–6488	arXiv:2305.14627	Verified
Retrieval-Augmented Generation — Lewis et al.	NeurIPS 2020	arXiv:2005.11401	Verified; page numbers unverified
Dense Passage Retrieval — Karpukhin et al.	EMNLP 2020, pp. 6769–6781	arXiv:2004.04906	Verified

Paper	Venue	Identifier	Status
Passage Re-ranking with BERT — Nogueira, Cho	**No venue found**	arXiv:1901.04085	Cite as preprint
BEIR — Thakur, Reimers, Rücklé, Srivastava, Gurevych	NeurIPS 2021 Datasets & Benchmarks	arXiv:2104.08663	Verified; title spelling differs between arXiv and proceedings
From RankNet to LambdaRank to LambdaMART — Burges	Microsoft Research TR, June 2010	MSR-TR-2010-82	Verified

Still to verify

- Read Liu et al. in full and fill in the methodology
- Confirm whether Nogueira & Cho was ever formally published
- Fetch proceedings page numbers for RAG and BEIR
- Pick one BEIR title spelling and use it consistently
- Confirm the GEO-bench Hugging Face path with the supervisor; the paper itself gives `generative-engines.com/GEO/`
- Run the post-2024 GEO novelty check